

Validation framework for sensor accuracy

Ensuring consistency and calibration across millions of embedded sensors, therefore, depends on a structured and validated production approach. A certified quality management system (such as ISO 13485) provides the foundation for rigorous process validation, focusing not only on final product inspection but on verifying that the entire manufacturing process consistently produces sensors with the intended characteristics. Quality control is implemented at multiple stages, including inline, online, and at-line monitoring, so deviations can be detected and corrected early rather than discovered at the end of production. Caring about the yield during all production steps is essential.

This process validation framework begins by defining the key sensor characteristics that must be optimised, such as mechanical sensitivity or temperature response. Established analytical methods are then applied to evaluate both the manufacturing process and the behaviour of the sensor itself.

Validation is structured into three main phases. Installation qualification (IQ) confirms that the system is correctly installed and meets required specifications such as temperature and pressure conditions. A key requirement at this stage is ensuring that all sensors are properly calibrated, even when they

Physical limits of nanoscale temperature sensors

- Temperature is fundamentally a statistical quantity and is only well-defined when a sufficiently large number of atoms are involved.
- At extreme miniaturisation (e.g., systems with only a few dozen atoms such as laser-cooled atomic traps), the concept of temperature begins to break down due to loss of standard statistical behaviour.
- Current nanoscale sensors remain far above this limit; even a $10\text{nm} \times 10\text{nm} \times 10\text{nm}$ sensor contains a large number of atoms, allowing temperature to be measured reliably using statistical principles.
- Shrinking sensors offer major benefits, including very low mass and extremely fast thermal response.
- Theoretically, response times could reach pico- to femtosecond scales; practical devices on substrates typically operate in the millisecond or sub-millisecond range.
- The main limitation is increased $1/f$ (flicker) noise as device size decreases.
- This noise is predominantly low-frequency and can often be mitigated using AC measurement techniques.

are embedded within the system and not directly accessible.

Output qualification (OQ) verifies that the process operates correctly once configured, with sensor outputs checked against expected specifications to ensure correct system behaviour under defined operating conditions.

Performance qualification (PQ) evaluates sensor performance in real-world applications, including measurements such as force, bending, or other operational stresses within the final system. In highly regulated industries such as medical device manufacturing, all three qualification stages are mandatory, while in less regulated industrial environments, greater reliance may be placed on traceability data such as material handling records, environmental conditions, and process timestamps,

particularly when supported by standards like ISO 9001.

Machine learning and artificial intelligence are used in some contexts to support the interpretation of sensor data by enhancing analytical workflows and improving model-based evaluation of complex signals. But the system is fundamentally grounded in structured validation and physical measurement principles rather than purely data-driven inference.

Nanosensor technology is changing the way we think about sensing by turning ordinary materials into 'smart' materials that can feel things like force, heat, and pressure on their own. Instead of attaching separate sensors, these systems are built directly into the material while still keeping its normal strength and flexibility. Although there are still challenges for novel technologies, such as complex manufacturing and industry adoption, new advances in automated production and better quality control are making it more realistic to use them widely. With strong testing methods to ensure accuracy and reliability, this technology could soon be widely adopted in many fields, such as cars, factories, and healthcare. **EFY**

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Material selection and its impact on sensor accuracy

- Material choice significantly affects sensor accuracy due to differences in how materials interact with the manufacturing process.
- Early development focused on silicon-based materials, including silicon oxide, silicon nitride, and silicon carbide, where additive manufacturing processes were already well established.
- Transitioning from silicon to polymers required substantial redevelopment because polymers behave very differently from semiconductor materials.
- Expansion into metals and glass introduced new challenges related to surface roughness, material thickness, and manufacturing residues such as oils or contaminants.
- These surface-related factors can directly impact sensor performance and fabrication quality, making cleaning and surface preparation critical steps.
- Over the past two years, processes have been refined to enable reliable integration across major material classes, including metals, ceramics, glasses, elastomers, and polymers.